# Q2: 131 + 134 Group

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2026-10-09

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## PSTAT 134 vs. non-PSTAT134

### Loading Data and Defining the ML Areas

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /Users/khoihuynh/Desktop/pstat197a/Module0
```

```
library(knitr)

background <- read_csv(here('data', 'raw', 'background-clean.csv'))
```

```
## Rows: 51 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): prog.prof, math.prof, stat.prof, upper_div_courses, courses, domain...
## dbl (4): response_id, prog.comf, math.comf, stat.comf
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
interest   <- read_csv(here('data', 'raw', 'interest-clean.csv'))
```

```
## Rows: 49 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): type, language, domains, areas
## dbl (1): response_id
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
joined     <- inner_join(background, interest, by = 'response_id')

ml_areas <- c(
  "Analysis or classification of images",
  "Deep learning and neural networks",
  "Data acquisition and engineering",
  "Natural language processing and analysis of text",
  "Predictive modeling, generally",
  "Statistical models and inference, generally"
)
```

We chose these six areas because they were the most closely related to machine learning. Students were able to select from eleven areas of interest, and most of them can be argued to connect to ML in some way. However, this selection focuses primarily on areas involving building, training, or applying predictive models, or preparing the data those models rely on. We recognize that this choice of slection is somewhat subjective and that a different selection could lead to slightly different results.

## PSTAT 134 Enrollment

Something we can look at first is how many students in our sample have taken PSTAT 134 compared to those who have not. This tells us how large each group is before we compare their maching learning interests.

```
q134 <- joined |>
  mutate(group_134 = if_else(str_detect(courses, fixed('PSTAT 134')),
                             'Took PSTAT 134', 'Did not take PSTAT 134'))

q134 |>
  count(group_134, name = 'n_students') |>
  mutate(proportion = n_students / sum(n_students)) |>
  kable(digits = 3, caption = 'Students by PSTAT 134 enrollment')
```

Table 1: Students by PSTAT 134 enrollment

| group_134 | n_students | proportion |
|---|---|---|
| Did not take PSTAT 134 | 36 | 0.735 |
| Took PSTAT 134 | 13 | 0.265 |

From this table, we can see that PSTAT 134 (Statistical Data Science) is not a commonly taken course among the students in our sample. Only 13 of the 49 respondents (26.5%) reported taking PSTAT 134, while 36

(73.5%) had not. This is a sharp contrast with PSTAT 131, which 44 of the 49 students (89.8%) had taken. Since PSTAT 131 is a prerequisite for PSTAT 134, most students have completed the introductory machine learning course but relatively few continued on to 134, leading to a unbalanced comparison.

## Measuring ML Interest for Each Student

Each student could select any number of the eleven interest areas, so we need a single measure of how interested each student is in machine learning. To do this, we can check which of our six ML areas each student selected and count them, giving every student an ML interest score from 0 to 6. This allows us to compare both groups on the scale!

```
area_long_134 <- q134 |>
  crossing(area = ml_areas) |>
  mutate(selected = str_detect(areas, fixed(area)))

ml_data_134 <- area_long_134 |>
  group_by(response_id, group_134) |>
  summarise(n_ml_areas = sum(selected), .groups = 'drop')

head(ml_data_134)
```

```
## # A tibble: 6 x 3
##   response_id group_134              n_ml_areas
##         <dbl> <chr>                      <int>
## 1        1001 Did not take PSTAT 134          3
## 2        1002 Did not take PSTAT 134          5
## 3        1003 Did not take PSTAT 134          6
## 4        1004 Did not take PSTAT 134          2
## 5        1005 Did not take PSTAT 134          6
## 6        1006 Took PSTAT 134                  3
```

```
ml_data_134 |>
  count(group_134, n_ml_areas) |>
  pivot_wider(names_from = n_ml_areas, values_from = n, values_fill = 0) |>
  kable(caption = 'Number of students by ML areas selected (out of 6)')
```

Table 2: Number of students by ML areas selected (out of 6)

| group_134 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Did not take PSTAT 134 | 1 | 3 | 7 | 12 | 10 | 3 |
| Took PSTAT 134 | 0 | 2 | 3 | 3 | 5 | 0 |

### Summary

```
ml_data_134 |>
  group_by(group_134) |>
  summarise(n = n(),
            mean_ml_areas = mean(n_ml_areas),
            median_ml_areas = median(n_ml_areas),
```

```
            min = min(n_ml_areas),
            max = max(n_ml_areas)) |>
  kable(digits = 2, caption = 'Number of ML-related areas selected, by PSTAT 134 enrollment')
```

Table 3: Number of ML-related areas selected, by PSTAT 134 enrollment

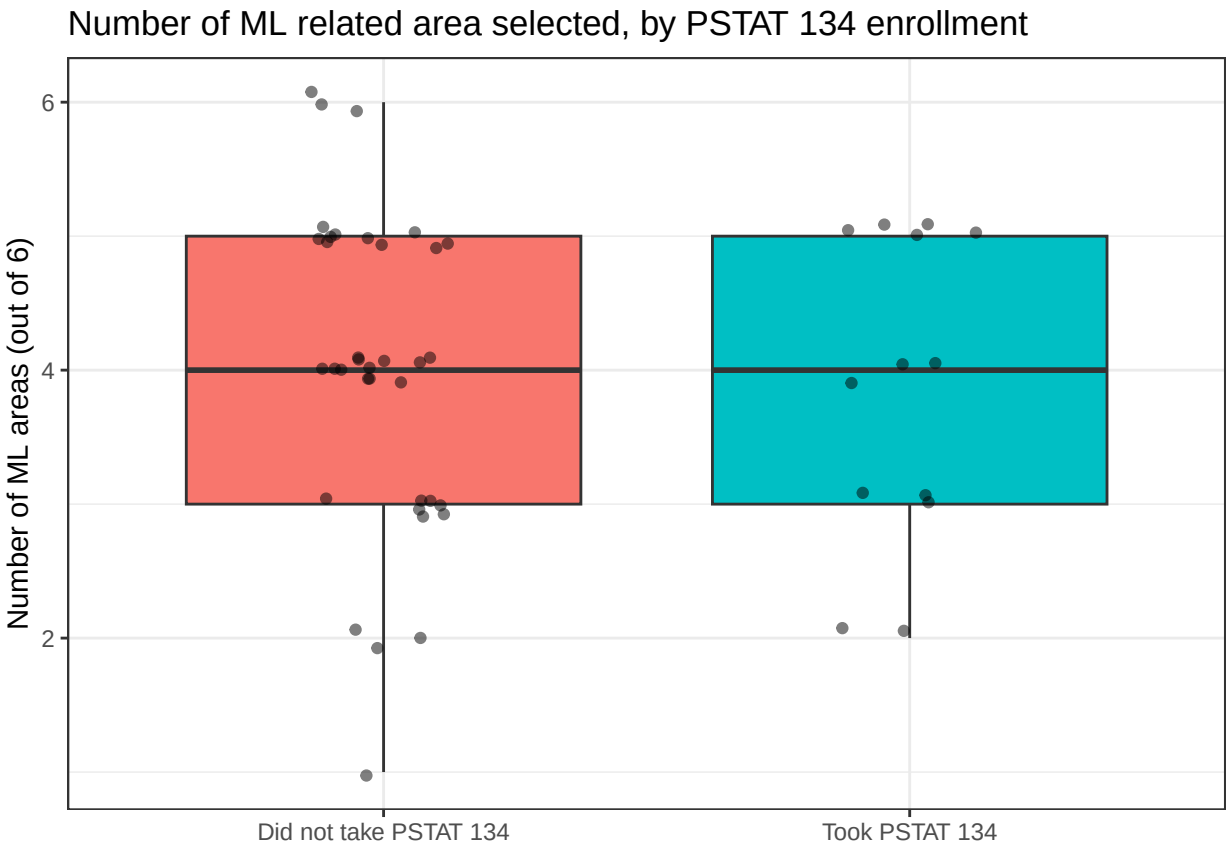
| group_134 | n | mean_ml_areas | median_ml_areas | min | max |
|---|---|---|---|---|---|
| Did not take PSTAT 134 | 36 | 4.00 | 4 | 1 | 6 |
| Took PSTAT 134 | 13 | 3.85 | 4 | 2 | 5 |

Students who did not take PSTAT 134 selected slightly more ML related areas on average (4.00) than those who took PSTAT 134 (3.85), although both groups share the same median of 4. Something to note is that students who did not take PSTAT 134 also had a wider range (1 to 6 areas) than those who took it to (2 to 5 areas), likely because that group is nearly three times larger. In all, we can see that the two groups show very similar levels of ML interest, suggesting that taking PSTAT 134 is not really associated with selecting more ML related areas.

## Distribution of ML Interest by Group

While the summary table gives us averages, it does not show how the scores are spread out within each group. A boxplot lets us compare the middle and range of ML interest scores side by side, and adding the individual points shows each student, which is especially helpful since the PSTAT 134 group only has 13 students.

```
ggplot(ml_data_134, aes(x = group_134, y = n_ml_areas, fill = group_134)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.15, height = 0.1, alpha = 0.5) +
  labs(title = "Number of ML related area selected, by PSTAT 134 enrollment",
       x = NULL, y = 'Number of ML areas (out of 6)') +
  theme_bw() +
  theme(legend.position = 'none')
```




The boxplots for both groups look nearly identical: each has a median of 4 ML areas, and the middle 50% of students in both groups selected between 3 and 5 areas. The main difference is in the range, since students who did not take PSTAT 134 spread from 1 to 6 areas, while those who took PSTAT 134 stayed between 2 and 5. Overall, the plot supports our summary table, showing that the two groups have very similar distributions of ML interest.

## ML Interest by Area

So far we have only looked at the total number of ML areas each student selected, which can hide differences in which areas each group prefers. Next, we can break the interest down by individual area and compare the proportion of students in each group who selected it. This shows whether students who took PSTAT 134 lean toward certain ML topics more than others.

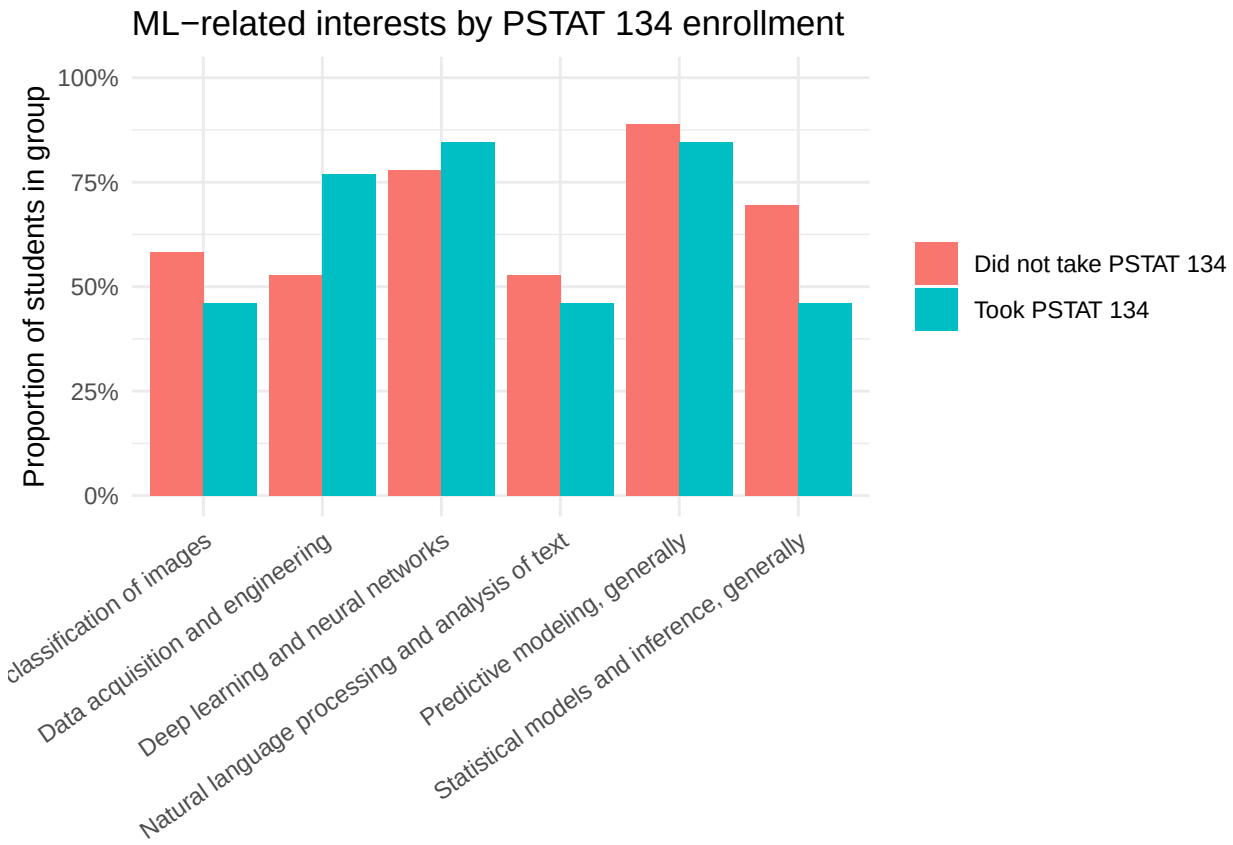
```
area_props_134 <- area_long_134 |>
  group_by(group_134, area) |>
  summarise(n_selected = sum(selected),
            n_students = n(),
            proportion = n_selected / n_students,
            .groups = 'drop')

area_props_134 |>
  select(area, group_134, proportion) |>
  pivot_wider(names_from = group_134, values_from = proportion) |>
  kable(digits = 3, caption = 'Proportion of students selecting each ML area, by PSTAT 134 enrollment')
```

Table 4: Proportion of students selecting each ML area, by PSTAT 134 enrollment

| area | Did not take PSTAT 134 | Took PSTAT 134 |
|---|---|---|
| Analysis or classification of images | 0.583 | 0.462 |
| Data acquisition and engineering | 0.528 | 0.769 |
| Deep learning and neural networks | 0.778 | 0.846 |
| Natural language processing and analysis of text | 0.528 | 0.462 |
| Predictive modeling, generally | 0.889 | 0.846 |
| Statistical models and inference, generally | 0.694 | 0.462 |

```
ggplot(area_props_134, aes(x = area, y = proportion, fill = group_134)) +
  geom_col(position = 'dodge') +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
  labs(title = 'ML-related interests by PSTAT 134 enrollment',
       x = NULL, y = 'Proportion of students in group', fill = NULL) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))
```




When we broke the ML interest down by area, some differences show up that the overall totals didnt catch initially. Students who took PSTAT 134 were a lot more into data acquisition and engineering (76.9% vs. 52.8%) and a little into deep learning (84.6% vs. 77.8%). On the other other hand, they picked statistical inference (46.2% vs. 69.4%). image analysis (46.2 % vs. 58.3%), and NLP (46.2% vs. 52.8%) less often. Predictive modeling was the top pick for both groups, with around 88.9% of non-134 students and 84/6% of 134 students choosing it. From this we can say that PSTAT 134 students might lean more toward data

engineering and away from traditional stats, but since there are only 13 of them, these differences could just be a random chance.

## Testing for Differences

The tables and plots show some difference between the two groups, but with only 13 students in the PSTAT 134 group, we should check whether these gaps are actually meaningful or could just be random noise. To check, we use Fisher's exact test for each area, since it works well with small groups, and a Wilcox on rank-sum test to compare the total number of ML areas picked, since those scores are counts rahter than normally distributed values.

```
# Fisher's exact test

area_long_134 |>
  group_by(area) |>
  summarise(took_134 = mean(selected[group_134 == 'Took PSTAT 134']),
            not_134 = mean(selected[group_134 == 'Did not take PSTAT 134']),
            p_value = fisher.test(table(group_134, selected))$p.value) |>
  kable(digits = 3, caption = "Fisher's exact test by ML area")
```

Table 5: Fisher's exact test by ML area

| area | took_134 | not_134 | p_value |
|---|---|---|---|
| Analysis or classification of images | 0.462 | 0.583 | 0.525 |
| Data acquisition and engineering | 0.769 | 0.528 | 0.191 |
| Deep learning and neural networks | 0.846 | 0.778 | 0.709 |
| Natural language processing and analysis of text | 0.462 | 0.528 | 0.754 |
| Predictive modeling, generally | 0.846 | 0.889 | 0.650 |
| Statistical models and inference, generally | 0.462 | 0.694 | 0.184 |

```
# Wiclxo rank sum test
wilcox.test(n_ml_areas ~ group_134,
            data = ml_data_134,
            exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  n_ml_areas by group_134
## W = 249.5, p-value = 0.7251
## alternative hypothesis: true location shift is not equal to 0
```

From the Fisher's test, it seems like none of the six ML areas showed a statistically significant difference between the two groups, since every p value was well above 0.05. Even with the biggest gaps, like data engineering ($p = 0.191$) and statistical inference ($p = 0.184$), were not strong enough to rule out random chance. Additionally, the Wilcox rank sum test found no significant difference in the total of number of ML areas picked between the two groups ($W = 249.5$, $p = 0.725$). Meaning that students who took PSTAT 134 and those who didn't showed pretty much the same overall level of interest in machine learning.

### 131 + 134 Group (WORK ON LATER)

## filter to students who took both 131 and 134

```
q2_134 <- joined %>%
  filter(str_detect(courses, fixed('PSTAT 131')),
         str_detect(courses, fixed('PSTAT 134')))
nrow(q2_134)
```

```
## [1] 13
```

## flagging ml areas

```
q2_134 <- q2_134 |>
  mutate(
    deep_learning = str_detect(areas, fixed('Deep learning and neural networks')),
    stat_inference = str_detect(areas, fixed('Statistical models and inference, generally')),
    images = str_detect(areas, fixed('Analysis or classification of images')),
    nlp = str_detect(areas, fixed('Natural language processing and analysis of text')),
    data_eng = str_detect(areas, fixed('Data acquisition and engineering')),
    predictive = str_detect(areas, fixed('Predictive modeling, generally')),
    ml_score = deep_learning + stat_inference + images + nlp + data_eng + predictive

  )
```

## Results

```
q2_134 |>
  summarise(n = n(),
            across(c(deep_learning, stat_inference, images, nlp, data_eng, predictive), sum),
            avg_ml_score = mean(ml_score))
```

```
## # A tibble: 1 x 8
##       n deep_learning stat_inference images   nlp data_eng predictive
##   <int>         <int>          <int>  <int> <int>    <int>      <int>
## 1    13            11              6      6     6       10         11
## # i 1 more variable: avg_ml_score <dbl>
```

```
q2_134 |>
  count(ml_score)
```

```
## # A tibble: 4 x 2
##   ml_score     n
##      <int> <int>
## 1        2     2
```

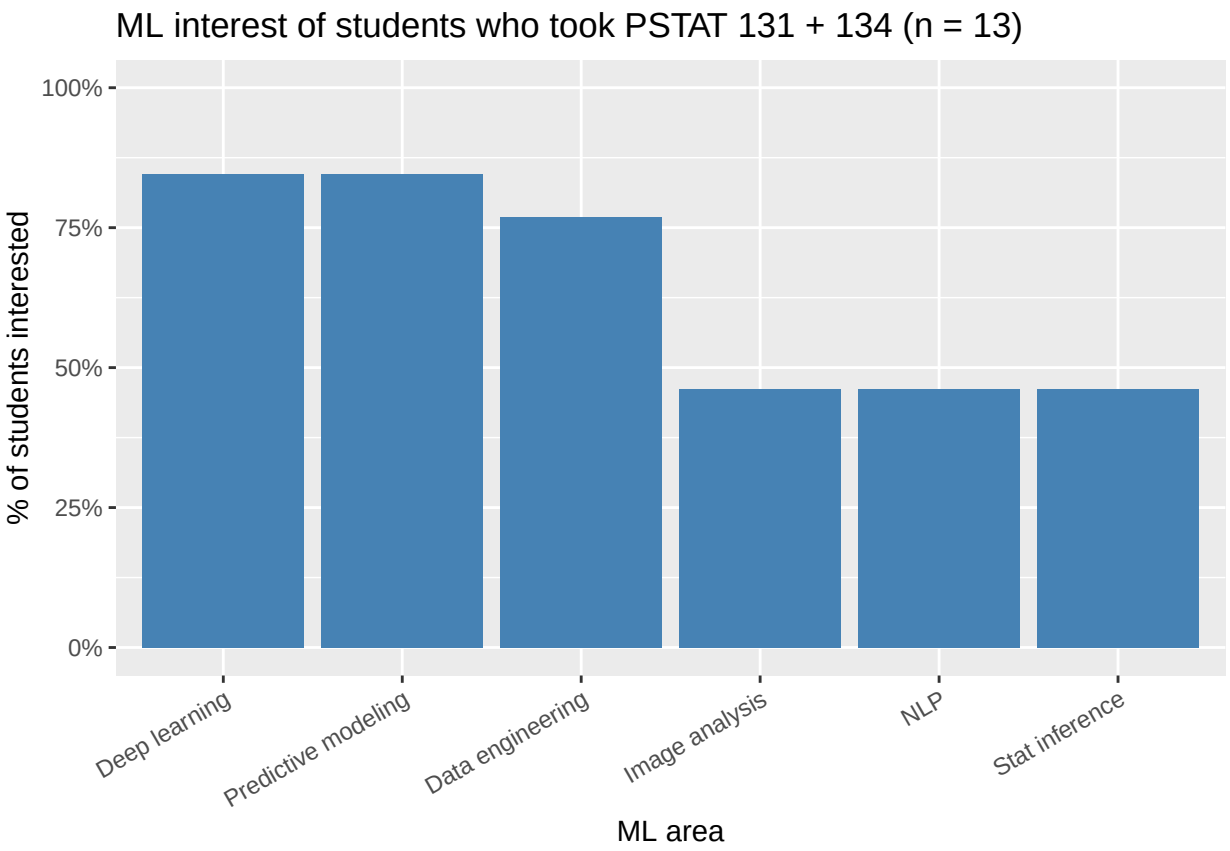
```
## 2        3     3
## 3        4     3
## 4        5     5
```

```
q2_134 |>
  count(language)
```

```
## # A tibble: 3 x 2
##   language          n
##   <chr>         <int>
## 1 No preference     4
## 2 Python            8
## 3 R                 1
```

# Plot

```
q2_134 |>
  summarise(`Deep learning` = mean(deep_learning),
            `Stat inference` = mean(stat_inference),
            `Image analysis` = mean(images),
            `NLP` = mean(nlp),
            `Data engineering` = mean(data_eng),
            `Predictive modeling` = mean(predictive)) %>%
  pivot_longer(everything(), names_to = 'area', values_to = 'pct') |>
  ggplot(aes(x = reorder(area, -pct), y = pct)) +
  geom_col(fill = 'steelblue') +
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
  labs(x = 'ML area', y = '% of students interested',
       title = 'ML interest of students who took PSTAT 131 + 134 (n = 13)') +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```




So out of the 13 students who took 131 and 145, the interest was highest in deep learning and predictive modeling (both having like 85%?).